

CareConnect: AI-Powered Companion for the Elderly

By

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ABSTRACT

This project is an AI-driven application development project for academic purposes. It aims to provide students with the methodology, concepts, and design of a mobile application tailored to the needs of elderly users. The project focuses on leveraging advanced AI technologies and machine learning techniques to address critical aspects of elderly care, including health management, cognitive engagement, and emergency response. The application will integrate multiple functionalities into a single platform, including medication and appointment management, emergency detection, gamified brain-stimulating activities, and caregiver connectivity. From a design perspective, emphasis is laid on the user-centric development process, ensuring the application is intuitive and accessible for elderly users. The AI design process involves integrating Natural Language Processing (NLP) for chatbot interactions, sensor-based algorithms for emergency detection, and image recognition technology for medication management. The development environment will involve the use of Android Studio for app development and Firebase for secure data storage. In addition to AI-powered features, gamification elements will be incorporated to promote cognitive engagement through interactive brain-stimulating games. Emergency response capabilities will include real-time detection of critical sounds and a visible offline emergency button for added safety. Notifications and updates will be shared with caregivers to ensure connectivity and support. Lastly, the application will serve as a comprehensive solution to enhance safety, independence, and quality of life for elderly users.

Area of Study (Minimum 1 and Maximum 2): **Mobile Application Development, Artificial Intelligence in Elderly Care**

Keywords (Minimum 5 and Maximum 10): **AI Chatbot, Emergency Detection, Cognitive Engagement, Medication Management, User-Centric Design, Caregiver Connectivity, Brain-Stimulating Games, Elderly Safety.**

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LIST OF ABBREVIATIONS

<i>AI</i>	Artificial Intelligence
<i>NLP</i>	Natural Language Processing
<i>HIPAA</i>	Health Insurance Portability and Accountability Act
<i>API</i>	Application Programming Interface
<i>DB</i>	Database
<i>IOT</i>	Internet of Things
<i>GDPR</i>	General Data Protection Regulation

CHAPTER 1 - Project Background

1.1 Introduction

The rapid aging of the world's population has accelerated the need for technological innovation to meet the special needs of older people [1]. The elderly population in Malaysia is defined as people aged 65 and above. From 2018 to 2019, the elderly population grew from 2.12 million to 2.21 million in less than a year. The population of Malaysians aged 65 and above is estimated to reach 7% in 2020. It is expected that the population will grow to 15% in 2040 [2]. As people are aging, it comes with increasing health risks and conditions [3]. Seniors often struggle with maintaining independence, adhering to medication schedules, remembering appointments, and staying socially engaged. At the same time, caregivers and families are seeking tools to ensure the safety and well-being of their elderly loved ones. These challenges underscore the need for innovative, technology-driven solutions that bridge the gap between caregiving and independence. The **Elderly Companion Application** has been conceptualized to meet this need.

The proposed application integrates advanced Artificial Intelligence (AI) technologies with user-friendly interfaces to create a holistic support system for seniors. By combining AI-powered tools such as chatbots, image recognition [4], cognitive engagement features, and emergency detection systems, this app addresses the core concerns of elderly care. The chatbot assists in managing medication schedules, recording medicine usage times, and providing reminders for medical appointments. Leveraging AI-based image recognition, the application ensures accuracy in tracking medication and appointments by analysing physical records, such as pill bottles or appointment cards. In addition, real-time notifications sent to caregivers enable them to stay informed and intervene if necessary.

Safety and emergency response are paramount features of the application. Using advanced AI, the app detects unusual sounds or events, such as shouting, phone dropping, or falls, and prompts the user to confirm their situation. If no response is detected, it automatically alerts designated caregivers or emergency services with the user's location. To ensure constant availability, an offline emergency button is

prominently displayed in the interface, allowing users to send alerts with ease, even in the absence of an internet connection.

1.2 Problem Statement

The increasing aging population worldwide has brought significant challenges in maintaining seniors' independence, safety, and mental well-being. This demographic shift has resulted in a growing number of older adults who require support in managing their daily lives. Seniors often face difficulties such as **remembering medication schedules, keeping track of medical appointments, and responding quickly to emergencies**. These challenges are further amplified when seniors live alone or apart from their family or loved ones, leaving them without immediate assistance when critical situations arise [5]. The absence of a close support network creates a vulnerability, making it harder for seniors to maintain a safe and healthy lifestyle.

Despite the availability of technological solutions, such as mobile applications designed to assist the elderly, most existing tools lack a holistic approach. Many of these applications focus on isolated features, such as medication tracking, emergency alerts, or appointment reminders, without integrating these functionalities into a single, cohesive platform [6]. This segmentation not only reduces the efficiency of these tools but also increases the complexity for elderly users, who may find it challenging to navigate multiple applications for different tasks. Additionally, many of these solutions fail to cater specifically to the unique needs of seniors, particularly in terms of simplicity in interface design, offline accessibility for emergencies, and features that support mental engagement, such as brain-training activities. Without addressing these critical usability factors, these tools often become impractical for their intended users, resulting in low adoption rates and underutilization of their potential benefits [7].

The lack of engaging features in current elderly care applications worsens social isolation and cognitive decline, often leading to memory loss or depression. Most tools focus on functional support, neglecting cognitive health and social connections, leaving both seniors and caregivers underserved. Caregivers, already burdened with

responsibilities, face challenges in ensuring consistent support, especially without integrated platforms for real-time monitoring and timely notifications.

There is a clear need for a comprehensive, technology-driven solution that combines medication management, emergency detection, and cognitive engagement into a single, user-friendly platform. Features such as intuitive interfaces, offline emergency access, and personalized tools can enhance usability and adoption.

Addressing these gaps not only improves seniors' quality of life but also reduces caregiver stress and aligns with public health goals of promoting active aging. An innovative, unified solution can empower seniors to live independently, enhance safety, and support caregivers, setting a new standard for elderly care technology.

1.3 Motivation

The aging of the global population poses unique challenges that require innovative solutions. Older adults often face difficulties in managing their daily lives, such as managing medications, making appointments, staying safe in emergencies. These challenges, such as managing cognitive decline and addressing social isolation, become even more daunting in the absence of sufficient caregivers, further complicating the support needed for the elderly. Despite growing awareness of all these issues, the lack of effective tools to comprehensively address these needs remains a significant gap in elder care.

Technology offers immense potential to transform the way care is provided to older adults. Mobile applications and artificial intelligence (AI) technologies have demonstrated their ability to simplify complex tasks and improve the quality of life for users. However, existing applications often fall short of integrating essential features into a single platform that caters specifically to the elderly. This fragmentation limits the effectiveness of current solutions, leaving seniors and their caregivers underserved.

The motivation for this project come from the need to **create a unified solution** that leverages advanced AI capabilities to provide holistic support for seniors. By

combining medication management, emergency detection, and engaging cognitive activities in one application, this project aims to bridge the gap in current elderly care solutions. It seeks to empower seniors to maintain their independence and well-being while reducing the burden on caregivers.

This project also aligns with the broader goals of promoting healthy aging and improving the quality of life for the elderly. By addressing their safety, cognitive health, and social engagement needs, the proposed solution aims to enhance the overall experience of aging. The development of this application is driven by a vision to not only solve immediate challenges but also set a new standard for technology-assisted elderly care in the future.

1.4 Contributions

Our analysis and design validate the feasibility and significance of the proposed Elderly Companion Application in addressing key challenges in elderly care. Firstly, the integration of AI-powered features such as medication tracking, emergency detection, and cognitive engagement confirms the effectiveness of a unified platform in **enhancing the independence and safety of seniors**. This provides a strong foundation for future research and development in comprehensive technology-driven elderly care solutions.

Secondly, the application's simplified interface and offline functionality address accessibility issues faced by seniors, especially those with limited technological proficiency or internet connectivity. These features ensure the **platform's usability and reliability across diverse user scenarios**.

Thirdly, the inclusion of interactive cognitive games and social engagement tools demonstrates the potential to combat cognitive decline and **reduce social isolation among seniors**. This aspect highlights the importance of holistic approaches in elderly care technology.

Finally, the real-time monitoring and notification system for caregivers **reduces the burden on family members** by providing timely updates and actionable insights into

CHAPTER 1

seniors' well-being. This feature enhances the efficiency and effectiveness of caregiving.

Overall, the proposed solution confirms the practicality and relevance of integrating advanced functionalities into a single platform, addressing current gaps in elderly care technology and contributing to the well-being of seniors and their caregivers.

CHAPTER 2

Literature Review

2.1 Previous Work on real life mobile applications

Real-life mobile applications play a vital role in addressing the healthcare and daily assistance needs of various demographics, particularly older adults. These applications utilize advanced technologies like AI and machine learning to enhance medication management, cognitive engagement, and social connectivity. The following sections review notable applications such as AgeWiser, Medisafe, and MyTherapy, focusing on their features, advantages, and limitations in providing comprehensive and accessible solutions for health and well-being.

2.1.1 AgeWiser

A real life application AgeWiser app exemplifies the integration of AI technology into senior care, addressing key challenges faced by older adults in maintaining independence, health, and social connectivity. The app incorporates a range of features [8] designed to improve physical, cognitive, and emotional well-being. Mobility support is provided through guided exercise videos, including Tai Chi and chair exercises, aimed at enhancing strength, balance, and fall prevention. Its AI-driven fall detection system monitors irregular movements to identify potential risks, enabling timely intervention and personalized care adjustments.

For cognitive health, AgeWiser includes brain games designed to boost memory and mental acuity. Additionally, the app features medication management tools that assist users in adhering to their prescribed schedules and detecting potential medication conflicts, ensuring better health outcomes. Social connectivity is a core focus, offering a platform for communication with family and friends, helping to combat loneliness and isolation. One of its standout features is the AI companion "Jen," an interactive avatar that provides personalized engagement, fostering a sense of companionship and support.

AgeWiser prioritizes user accessibility with a simple, intuitive interface designed for older adults. Available on Android and iOS platforms, the app ensures data privacy and security by encrypting user information during transmission. Despite its innovative features, further research could explore its real-world efficacy in enhancing the quality of life for diverse elderly populations and addressing specific limitations like technological familiarity among older adults. Overall, AgeWiser demonstrates the potential of AI-powered applications in transforming elderly care through comprehensive and user-friendly solutions.



Figure 2.1.1 AgeWiser Application: Mobility



Figure 2.1.2 AgeWiser Application: Brain game & Pill Mangement

2.1.2 Medidafe

Medisafe is a widely recognized medication management application designed to enhance adherence to prescribed regimens and support overall health tracking [9]. It is particularly beneficial for individuals managing complex medication schedules, as it offers customizable reminders and detailed dosing schedules. The app integrates advanced features such as drug interaction warnings, which notify users of potential conflicts between medications, ensuring safe usage. Additionally, Medisafe includes a caregiver support system called "Medfriends," allowing family members or caregivers to receive notifications when a dose is missed, fostering a collaborative approach to medication adherence.

The application also supports health tracking by monitoring metrics like blood pressure, glucose levels, and weight, enabling users to gain a holistic view of their health. Medisafe emphasizes data security, adhering to HIPAA standards, and employs 256-bit encryption to protect sensitive user information [10]. Its user-friendly interface includes visual representations of medications, such as pill images and clear scheduling, which simplify navigation and usage for all demographics, particularly older adults [10].

While Medisafe offers comprehensive features, some limitations exist. The absence of web-based access limits its usability to mobile devices. Despite these drawbacks, Medisafe demonstrates significant potential as an accessible and effective tool for medication management. Its integration of reminders, health tracking, and caregiver support addresses critical challenges in medication adherence, making it an asset in personal and clinical healthcare management. Future improvements could focus on expanding accessibility and introducing features for a broader range of health conditions.

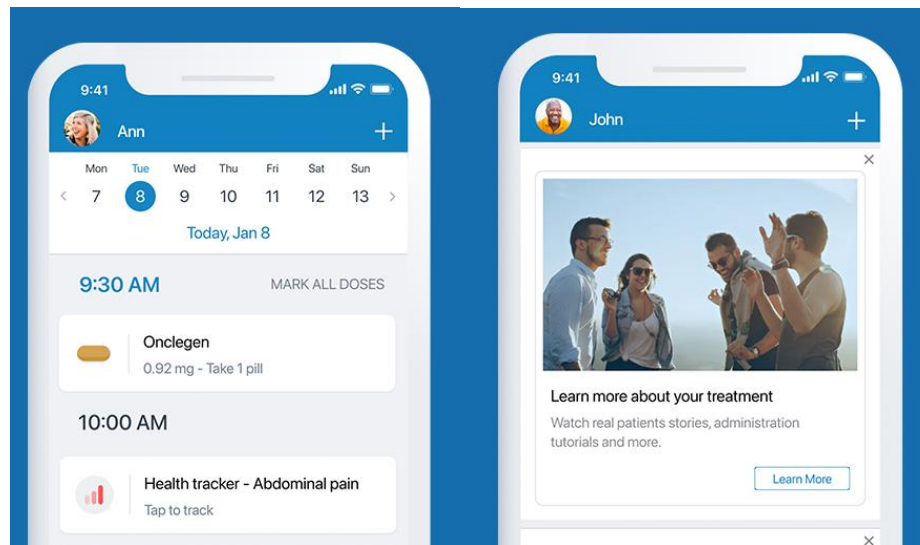


Figure 2.1.3 Medisafe Application

2.1.3 MyTherapy

The MyTherapy app is a comprehensive health management tool designed to support medication adherence and overall well-being [11]. It combines medication reminders with features for tracking symptoms, monitoring health metrics such as blood pressure, glucose levels, and weight, and logging physical activities [12]. By offering personalized reminders, the app ensures timely adherence to prescribed medication schedules, catering to users managing complex regimens. Additionally, the app generates detailed health reports based on logged data, which can be shared with healthcare providers to facilitate more informed clinical consultations.

One of MyTherapy's strengths is its user-friendly interface [12], which is designed to be accessible to individuals across different age groups. It supports multiple languages and includes accessibility features like Android's TalkBack and iOS's VoiceOver, ensuring inclusivity for visually impaired users. The app also prioritizes data privacy, complying with stringent European regulations to safeguard personal information and preventing unauthorized third-party data access.

In addition to the previously mentioned limitations, some users have reported that the MyTherapy app may experience notification issues, particularly when battery-saving modes are enabled on their devices [13]. This can result in reminders not functioning as intended, potentially affecting medication adherence.

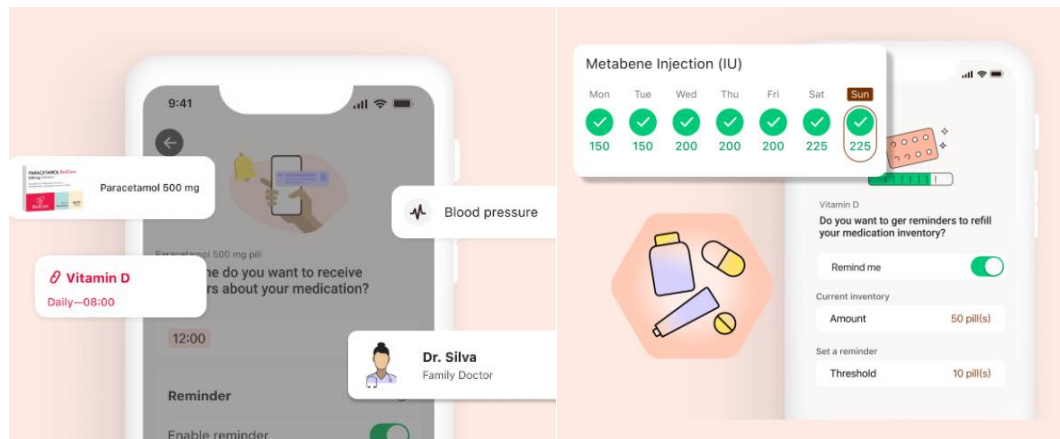


Figure 2.1.4 MyTherapy Application

Table 2.1 Benchmark Between Three Mobile Applications

Application	App Features	Advantages	Limitation
AgeWiser [8]	- Fall prevention - Cognitive games - Medication tracking - AI companion "Jen" - Social connectivity	- User-friendly interface - Drug interaction alerts - Caregiver integration - Adheres to strict privacy laws	- Lacks offline access - Initial setup may be tedious for some users
Medisafe [9]	- Medication reminders - Caregiver notifications (Medfriends) - Drug interaction warnings - Health tracking	- Comprehensive health management - Accessible interface - Multiple language support - Shareable health reports	- Time-consuming setup for extensive medications - Occasional notification issues with battery-saving modes

MyTheraphy [11]	- Medication reminders - Health tracking (blood pressure, glucose, weight) - Symptom tracking - Activity logging	- Combines mobility, cognitive health, and social engagement - Real-time fall detection - Personalized care	- Some features may require constant internet access for optimal performance
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2.2 Previous Work on Chatbot/ Voice Assistant

Chatbots and voice assistants have become essential tools for improving interaction between humans and machines. They use technologies like Natural Language Processing (NLP) and speech recognition to perform tasks, provide information, and enhance accessibility. Various studies have explored their applications in fields such as healthcare, education, and daily assistance, highlighting their potential and areas for improvement.

2.2.1 NLP based AI Voice Assistant

In [14] the "NLP-based AI Voice Assistant" paper highlights the role of AI systems in human-machine interaction, focusing on voice assistants and their applications. It references a variety of methodologies, such as acoustic analysis, decision tree algorithms, and NLP techniques, while also discussing practical uses like accessibility for visually impaired users and automation of daily tasks. However, the review primarily lists studies without critical comparison or analysis. It does not explicitly identify gaps in the literature, and the discussion is fragmented, lacking synthesis of findings. Challenges like data bias and variability in accents are mentioned indirectly but not explored in depth. Overall, while the review provides technical insights, it could benefit from a clearer structure, critical evaluation of prior work, and identification of research gaps to better justify the proposed system.

2.2.2 Artificial Intelligence-based Voice Assistant

IN [15], the authors have proposed an AI-based voice assistant designed to perform a variety of user commands using Python and Google Text-to-Speech (gTTS). The voice assistant captures user inputs through a microphone, processes them into text, and executes specific tasks such as playing music, retrieving weather updates, conducting Google searches, taking screenshots, and translating text into different languages. The processed responses are then converted into audio using the Playsound library to provide seamless interaction.

The core of the proposed system relies on **Automatic Speech Recognition (ASR)**, which incorporates Acoustic Analysis to process speech waveforms. The ASR breaks down speech into three modelling levels: **acoustic modelling** to identify speech elements, pronunciation modelling to account for variations like accents, and language modelling to establish contextual probabilities. This layered approach enables the assistant to handle diverse linguistic inputs and provide accurate responses.

The system includes additional features like customizable assistant names, session-specific memory, and emergency alert capabilities, making it adaptable to various user needs. The authors also highlight the assistant's ability to automate repetitive tasks and integrate with everyday applications, enhancing accessibility and convenience. Implemented using Python's flexible programming environment, the assistant is scalable for further enhancements, such as improved dialogue recognition and task automation.



Figure 2.2.1 Process of ASR

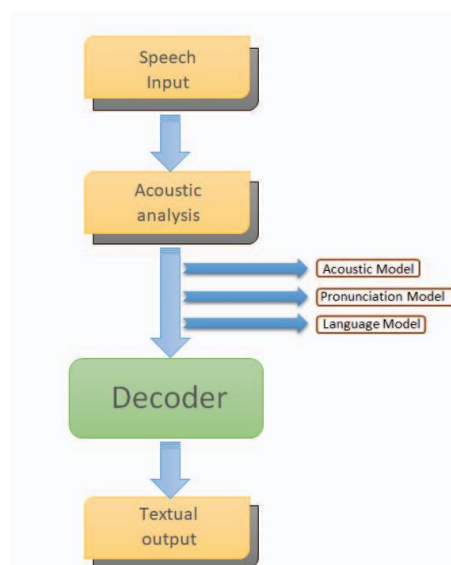


Figure 2.2.2 Acoustic Analysis

2.2.3 Enhancing College Chat Bot Assistant with the Help of Richer Human Computer Interaction and Speech Recognition

Chatbots have gained significant attention over the last decade, being widely adopted across various domains such as e-commerce platforms, college websites, and customer service. An advanced college chatbot has designed to address the challenges faced by students and parents during the admission process [16]. The chatbot supports both text and voice interaction, offering a more engaging and user-friendly experience. It aims to reduce manual effort, eliminate queues at help desks, and provide 24/7 support for admission-related queries.

The authors review previous work in chatbot development, highlighting pattern-matching approaches using relational databases, stemming techniques for keyword extraction, and machine learning models for natural language understanding. Examples include Bayu Setiaji and Ferry Wahyu Wibowo's database-driven chatbot and Dungeon Lee et al.'s emotionally responsive chatbot for counselling [11]. However, the authors note that existing chatbots often fail to bridge the gap between human-like interactions and system responses, leading to suboptimal user experiences.

The proposed system leverages Natural Language Processing (NLP) and Automatic Speech Recognition (ASR) for improved conversational abilities. It employs MongoDB for backend storage and RESTful APIs for robust communication. The chatbot classifies queries into FAQs and transactional questions, dynamically adapting responses based on context. Features like error correction, fallback questions, and multilingual capabilities enhance its usability.

Despite its innovative design, the chatbot faces challenges in handling ambiguous queries, misinterpretation of commands, and grammar errors. The authors emphasize the need for advanced NLP techniques and suggest future enhancements such as multilingual support and local language integration to make the system more inclusive and realistic. This research contributes to the growing field of intelligent virtual assistants, emphasizing the importance of seamless human-computer interaction.

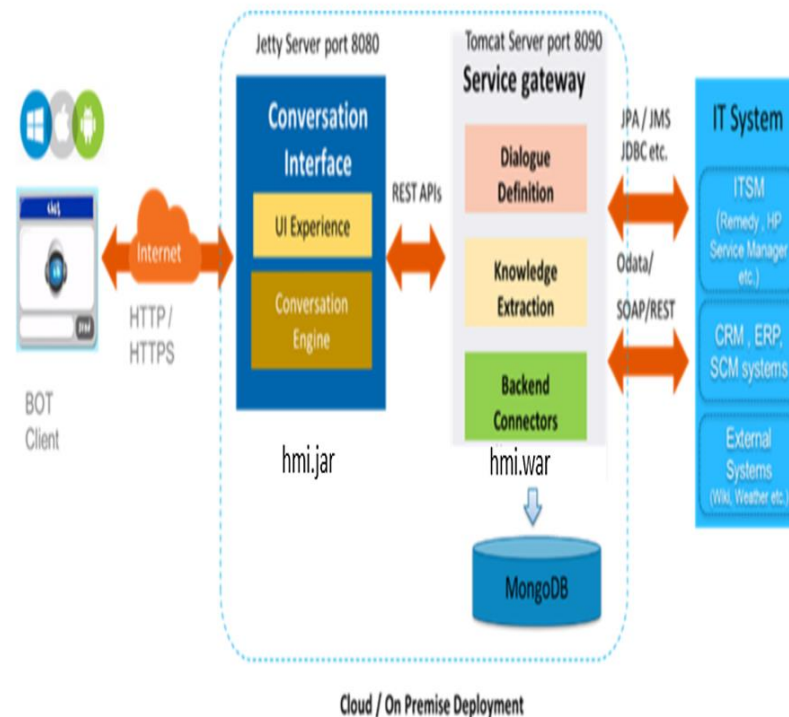


Figure 2.2.3 Cloud/ On premise Deployment Architecture

2.2.4 Chatbot Using A Knowledge in Database

The authors in [17] propose a chatbot framework that utilizes relational database management systems (RDBMS) to manage knowledge for human-to-machine conversations. The authors describe a pattern-matching approach where user input sentences are compared with stored patterns in the database, and responses are generated based on similarity scores. The bigram model is used for similarity calculations, dividing sentences into overlapping two-character sequences to handle misspellings and syntax errors effectively.

The chatbot's core relies on a database storing patterns, templates, and conversation logs, with normalization functions to correct misspellings and punctuation errors. The authors implemented the chatbot interface using Pascal and Java, enabling a user-friendly interaction platform. SQL-based stored procedures and functions are used for pattern matching, ensuring easy integration and customization.

The authors highlight the practical applications of the chatbot in handling natural language queries, particularly in Indonesian, and emphasize the flexibility of the RDBMS-based approach for updating knowledge and supporting multi-language implementation.

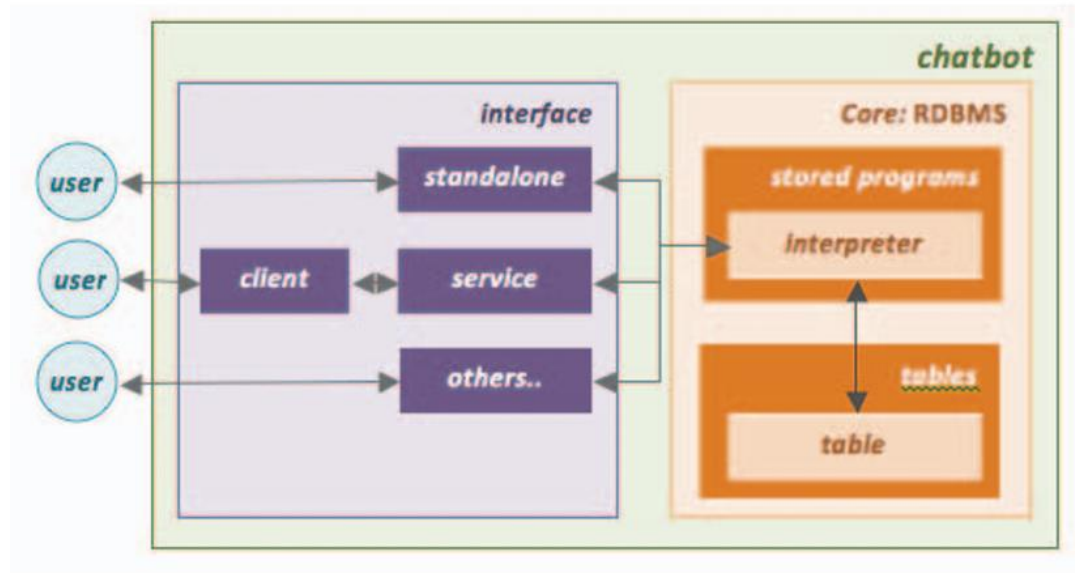


Figure 2.2.4 Global Design of Chatbot

2.2.5 Benefits of, Barriers to, and Needs for an Artificial Intelligence–Powered Medication Information Voice Chatbot for Older Adults: Interview Study With Geriatrics Experts

The author propose an AI-powered medication information voice chatbot to assist older adults in managing their complex medication regimens [18]. The authors emphasize that the chatbot leverages voice-based technology to address common challenges faced by older adults, such as diminished vision, reduced dexterity, and cognitive burdens. By providing features like medication reminders, instructions, and side effect information, the chatbot aims to improve adherence and support overall health. It also facilitates communication by connecting users with pharmacies, healthcare providers, and caregivers, thereby reducing the dependency on manual record-keeping and enhancing medication management. The authors highlight the chatbot's potential to

make health management more accessible through its ease of use and voice-based interaction.

Despite its advantages, the authors identify barriers to the adoption of this technology, particularly among older adults aged 75 and above or those from lower socioeconomic backgrounds, who may struggle with technology familiarity. Privacy and security concerns also emerged as significant issues, as some older adults are wary of data misuse or breaches. Additionally, usability challenges, such as complex interfaces, troubleshooting difficulties, and voice recognition accuracy, were noted as areas needing refinement. The authors suggest that these barriers could be mitigated through the design of an affordable, user-friendly chatbot with clear language, native language support, and technical assistance.

The author further proposes that integrating the chatbot with existing technologies, such as smart speakers or patient portals, could enhance its adoption and usability. They argue that the system should be designed to cater to diverse user needs, including personalized interactions and seamless integration with caregivers and healthcare providers. The proposed chatbot represents a significant step toward improving medication adherence and overall health management for older adults, but the authors stress the importance of addressing barriers like cost, privacy, and usability to achieve widespread adoption. Future research should involve diverse populations and real-world testing to validate the chatbot's effectiveness and refine its design.

2.2.6 Assisting Personalized Healthcare of Elderly People: Developing a Rule-Based Virtual Caregiver System Using Mobile Chatbot

This [19] article focuses on creating a virtual caregiver system to address the personalized healthcare needs of elderly individuals. The proposed system integrates a rule-based "mind monitoring" service with mobile chatbot technology to encourage dialogue-based interaction, allowing elderly users to express their mental and physical states. Unlike traditional health monitoring approaches that rely on external sensors, this system prioritizes internal expressions such as emotions, thoughts, and feelings.

The chatbot engages users by asking daily questions, which are answered through simple "yes" or "no" responses or via speech recognition. These responses are

quantified, visualized as graphs, and used to generate personalized summaries and advice. The aim is to enhance self-reflection and promote better self-care among elderly users. The study demonstrated the system's feasibility through a 14-month experiment involving elderly and younger participants, showing a response rate above 80% and reflecting real-life events like stress from COVID-19 restrictions.

Despite its effectiveness in usability and interaction, the authors highlight challenges such as the lack of gamification and the repetitive nature of questions, which could lead to boredom. They suggest integrating sensor-based technologies and gamification to improve engagement and functionality. The study underscores the potential of mobile chatbots as virtual caregivers, emphasizing their cost-effectiveness and scalability compared to physical care robots.

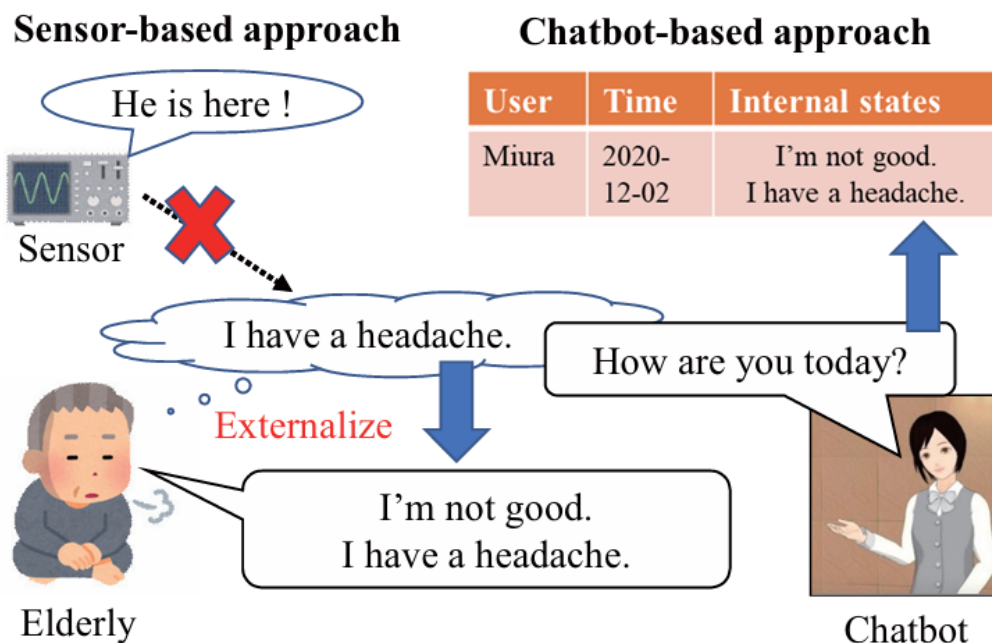


Figure 2.2.5 Sensor-based And Chatbot-based Approach For Elderly People

2.3 Previous Work on Elderly Tracking And Emergency Detection Mobile Applications

Elderly tracking and emergency detection applications use advanced technologies like accelerometers, machine learning, and real-time monitoring to ensure the safety and well-being of older adults. These systems focus on detecting critical events, such as falls, and providing timely alerts to caregivers, improving the quality of life for elderly individuals. The following sections summarize notable advancements, practical applications, and limitations in this domain.

2.3.1 Mobile applications for elderly healthcare: A systematic mapping

The study categorizes mobile health interventions using the World Health Organization's (WHO) Digital Health Initiatives framework, finding that most applications focus on Personal Health Tracking for independent elderly users. These apps emphasize monitoring vital signs, medication adherence, and scheduling medical appointments [20]. Despite advancements, the study identifies a significant gap in applications targeting dependent elderly users and public health system management, revealing an opportunity for research and innovation.

The analysis also highlights the spatiotemporal distribution of research, showing that most studies originate from developed countries, with the United States leading the publication count. In terms of validation techniques, controlled experiments and usability evaluations dominate, reflecting a focus on ensuring app effectiveness and user satisfaction. However, the study notes a lack of robust evaluation methods in many papers, leaving room for methodological improvement.

2.3.2 Development of Smart Elderly Care Mobile Application for Health Management System

Nowadays, elderly care faces numerous challenges, particularly in managing health effectively for both the elderly and their caregivers. Mobile applications like SECA are designed to address these health management challenges, especially in Malaysia, where the elderly population is rapidly increasing, necessitating systematic solutions to

overcome inefficiencies in traditional methods [21]. SECA simplifies and streamlines health management with five core features: Medical Appointment, Medication, Daily Health Data, Health Summary, and Elderly Profile. These features allow users to manage appointments, track medication, monitor health parameters, and maintain comprehensive health records efficiently.

The application employs Firebase for secure and stable data storage, ensuring information remains accessible even in case of device malfunctions. Built using Android Studio, SECA minimizes development and maintenance costs, offering a cost-effective solution for elderly care. The authors emphasize the app's user-friendly design, making it intuitive and accessible for both elderly users and their caregivers. Additional features, such as a COVID-19 vaccination record and personal profile management, further enhance usability.

The study acknowledges limitations in existing mobile health applications like Pill Monitor and BookDoc, which suffer from instability, restricted usability, and narrow scopes. SECA addresses these gaps by offering a comprehensive suite of features tailored to elderly health management needs. The authors propose future improvements, such as multi-language support, healthcare facility directories, and media upload options, to make the app more versatile and inclusive. SECA demonstrates the transformative potential of mobile technology to revolutionize elderly care by improving health management, fostering independence, and enhancing caregiver coordination.

2.3.3 Design of a Wearable Healthcare Emergency Detection Device for Elder Persons

Due to the high prevalence of falls as a leading cause of severe injuries and mortality among older adults, particularly those with dementia or metabolic disorders, the authors introduce a wearable system for detecting falls to improve safety and quality of life [22]. The proposed system employs a neural network embedded on an Arduino Nano 33 BLE Sense board for real-time movement detection, recognizing activities such as walking, running, and falling. Alerts are sent via Bluetooth to a connected smartphone, which notifies healthcare professionals if no response is received within 60 seconds,

including the user's location and activity data to enhance emergency response efficiency.

The wearable device is portable, low-cost, and suitable for both indoor and outdoor use. Designed as a single wrist-mounted device, it minimizes complexity and enhances user comfort. The study validates its performance using pre-existing datasets and real-world testing, achieving an activity recognition accuracy of 78%. Battery consumption is reduced by processing data locally on the device instead of streaming it to external systems.

Despite its potential, the authors acknowledge limitations, including the system's modest accuracy improvement compared to chest-mounted sensors and the simplified neural network model that may restrict its ability to classify complex activities. Future directions involve expanding detectable activities, integrating speech recognition for comprehensive emergency detection, and applying advanced privacy and security measures. This work demonstrates a significant step forward in developing accessible and efficient fall detection technologies for elderly care.

2.3.4 Fall Detection Using Accelerometer-Based Smartphones: Where Do We Go From Here?

Falls are the second leading cause of unintentional injury deaths worldwide, according to the World Health Organization [22]. More than 37 million falls each year are severe enough to require medical attention [22]. The author reviews solutions for fall detection using smartphone accelerometers, addressing falls as a critical health issue among older adults [23]. The study screened 267 articles, focusing on 15 that utilized either threshold-based algorithms or machine learning techniques for fall detection. Machine learning methods included decision trees, k-nearest neighbors, boosting, and neural networks, demonstrating sensitivity ranging from 60.4% to 99.3% and specificity from 74.6% to 100%.

The authors identify challenges in developing fall detection systems, such as limited participant diversity, small datasets, and artificial experimental setups that may not reflect real-world conditions. Most studies rely on simulated falls by young individuals, raising concerns about the algorithms' validity for elderly users. The review also

highlights gaps in evaluation methods, including the lack of standardized testing and real-world validation. To address these issues, the authors recommend expanding the scope of studies, improving data science methodologies, and adopting advanced machine learning techniques like deep learning.

This research emphasizes the potential of smartphone-based fall detection applications as affordable and accessible tools to enhance elderly safety. It calls for the inclusion of more diverse datasets, rigorous validation processes, and sophisticated algorithms to improve the reliability and practicality of these systems in real-world settings.

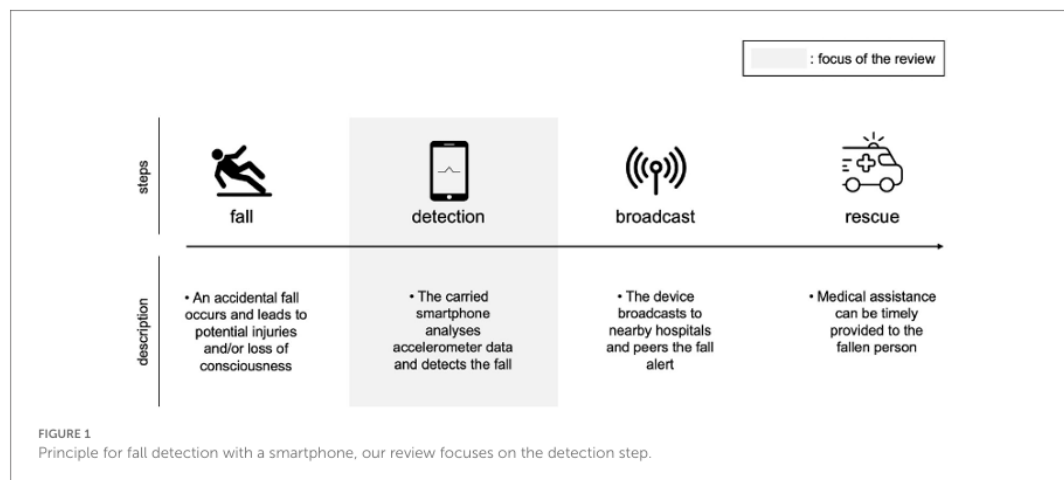


Figure 2.3.1 Principle For Fall Detection

2.3.5 The Application of Machine Learning on the Sensors of Smartphones to Detect Falls in Real-Time

Falls are a significant global public health issue, ranking as the second leading cause of accidental fatalities after motor accidents, with approximately 684,000 deaths annually. Over 80% of these fatal falls occur in developing and middle-income nations, with individuals over 60 experiencing the highest death rates. Even non-fatal falls are a major concern, with more than 37.3 million falls each year severe enough to require medical attention [24]. To address this issue, the study [25] focuses on developing a smartphone-based fall detection system leveraging machine learning to enhance safety for the elderly. By utilizing accelerometer sensors embedded in smartphones, the proposed system provides real-time monitoring to categorize activities as "fall" or "not

fall" using machine learning classifiers like Support Vector Machines (SVM) and k-Nearest Neighbors (KNN).

Data from smartphone accelerometers, recording x, y, and z-axis activity, is processed using MATLAB to identify patterns indicative of falls. A web-based platform connects the smartphone to MATLAB for real-time monitoring and emergency notifications, offering an efficient method for fall detection. The system's dataset includes diverse activities and fall scenarios, ensuring comprehensive training and testing. Results highlight the superior performance of the KNN model, achieving over 99.6% accuracy, sensitivity, specificity, and F1-score, with fewer misclassifications compared to SVM. The study emphasizes the system's effectiveness as a cost-effective and accessible solution, particularly beneficial for elderly individuals living independently.

Despite its potential, the authors recognize limitations, such as high energy consumption and the need for continuous signal processing. They propose optimizing energy efficiency by reducing data collection during inactivity. This work contributes to the field of wearable technology, showcasing the potential of machine learning-driven solutions to improve elderly care and safety.

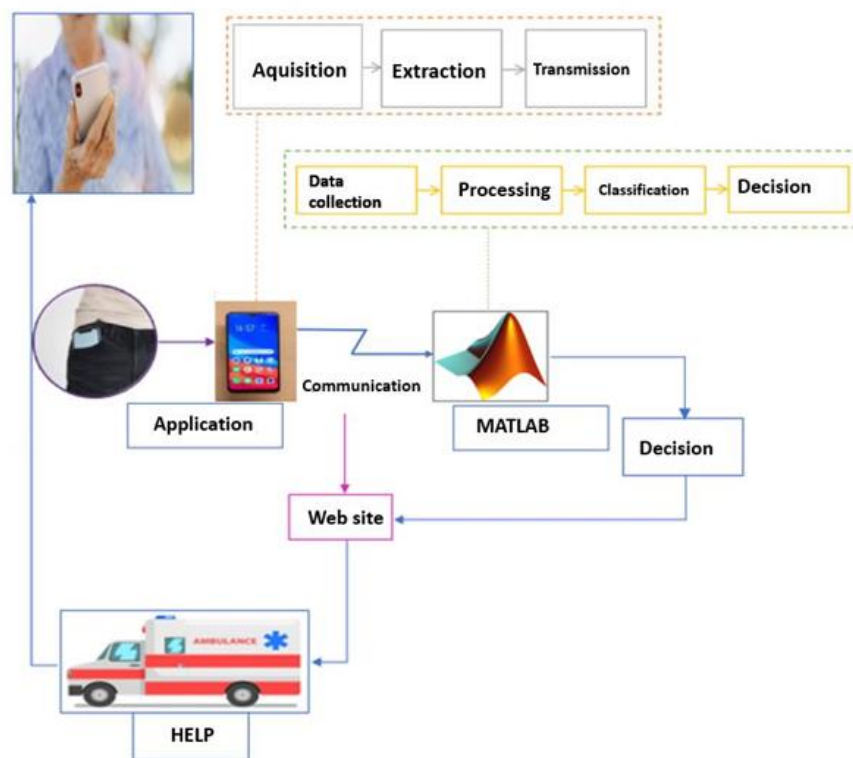


Figure 2.3.2 Falls Remote Monitoring System architecture

2.4 Previous Work on Cognitive Engagement

Cognitive engagement is essential for maintaining mental health, memory, and overall well-being, particularly among elderly individuals. Gamification has emerged as an effective tool to foster cognitive engagement by integrating game elements such as rewards, progress tracking, and interactive challenges into digital applications. These technologies aim to enhance user motivation, memory retention, problem-solving skills, and social interaction. The following sections explore notable studies and applications that leverage gamification to address cognitive decline and promote mental health, focusing on their features, benefits, and limitations.

2.4.1 A systematic review of gamification in e-Health

Gamification has emerged as a promising tool to enhance cognitive health and engagement in various domains, including e-Health. [26] conducted a systematic review on gamification in e-Health and highlighted its potential to improve user motivation, cognitive engagement, and social interaction. The study emphasizes that gamified applications integrate game mechanics such as leaderboards, badges, and progress tracking to provide users with a sense of achievement and competition. These elements encourage users to stay engaged and complete tasks with increased enthusiasm. For example, gamification has been successfully employed in health behaviour change systems to promote adherence to treatment plans and encourage cognitive development through brain-training exercises.

Further evidence indicates that gamified tools can effectively engage elderly users by incorporating personalized rewards, interactive tasks, and visual storytelling. Cognitive training games, often used in serious health applications, are tailored to improve memory, attention, and processing speed, crucial for elderly users facing cognitive decline. These applications use dynamic feedback and levelling systems to provide an adaptive gaming experience that matches the user's cognitive abilities.

However, author also highlight limitations in current gamified e-Health applications. While these tools show short-term success in improving engagement, sustaining motivation over the long term remains a challenge. To address this, the integration of user-centered design and the involvement of healthcare professionals in the

development process are suggested to ensure the applications meet the diverse needs of the elderly population.

This review underscores the importance of leveraging gamified applications as a medium for cognitive engagement and health management in elderly care. By combining gamification with advanced AI features and a user-centric approach, these applications can significantly enhance the mental well-being and social interaction of elderly users.

2.4.2 Gamification in Apps and Technologies for Improving Mental Health and Well-Being: Systematic Review

The author in [27] states that gamification is widely used in mental health interventions to enhance user engagement and outcomes. By incorporating game elements such as levels, rewards, progress tracking, and narrative themes, gamification fosters a sense of achievement and motivation among users. The review identifies that these elements are primarily applied to mental health domains like stress management, anxiety reduction, and mood enhancement.

However, the author highlights several challenges associated with gamification. One significant limitation is the lack of integration with robust theoretical frameworks, particularly those related to behaviour change. Many applications rely heavily on extrinsic motivators, such as rewards and leaderboards, while intrinsic motivators, like autonomy and meaningfulness, are underutilized. This gap potentially limits the long-term sustainability of user engagement and impact on mental health outcomes.

The author also points out issues with inconsistent terminology and evaluation methods in the field, which complicates the comparison and generalization of findings. Furthermore, many gamified interventions narrowly focus on mental health, missing opportunities for broader applications, such as enhancing cognitive skills or social interactions. The underutilization of evidence-based behavior change techniques further limits the effectiveness of these interventions.

Despite these challenges, the author emphasizes that gamification holds promise for mental health interventions. They propose integrating gamification with established

behavior change frameworks and enhancing intrinsic motivators to create more effective and sustainable applications. Additionally, they advocate for rigorous methodologies to evaluate the long-term impact of gamified interventions, ensuring that they contribute meaningfully to mental health and overall well-being.

2.4.3 Creative Brain Training Apps and Games Can Help Improve Memory, Cognitive Abilities, and Promote Good Mental Health for The Elderly

In [28], author explores the role of creative brain training apps and games in improving cognitive abilities, memory, and mental health among the elderly. These applications are designed to enhance memory retention, problem-solving, and attention through engaging and interactive activities. Examples include memory games, pattern recognition tasks, and adaptive difficulty exercises tailored to the abilities of elderly users. By incorporating gamified elements such as progress tracking, rewards, and competition, these apps aim to motivate users and provide a fun, interactive experience.

The study emphasizes the importance of accessibility, highlighting features like user-friendly interfaces, simplified gameplay mechanics, and personalized training modules. These design elements ensure that elderly individuals, who may have limited experience with technology, can effectively engage with the applications. Additionally, the incorporation of social interaction features, such as multiplayer modes and family-friendly activities, promotes intergenerational bonding and reduces social isolation among the elderly.

Despite these advantages, the study notes challenges in developing brain training applications for older adults. These include balancing game complexity to avoid frustration, ensuring long-term user engagement, and addressing technical accessibility for individuals unfamiliar with digital platforms. The effectiveness of gamified cognitive training remains a topic of debate, with further research needed to establish its long-term cognitive and emotional benefits.

This research underscores the potential of gamified brain training apps to enhance cognitive health and social well-being for elderly individuals. By integrating tailored challenges, gamified features, and user-friendly designs, these applications can provide meaningful and enjoyable cognitive stimulation for aging populations.

2.5 Limitation of Previous Studies

Previous studies highlight various limitations in existing systems and proposed solutions. The limitation of AgeWiser app [8] include the potential dependence on constant internet access for optimal functionality of certain features. These limitations highlight areas for improvement in each app to better cater to user needs and provide a more seamless experience. Next, the limitations of the Medisafe app [9] include the lack of offline access, and a potentially tedious initial setup process for users with extensive medication regimens. Last, the MyTherapy app [11] faces limitations such as the time-consuming setup process for entering a large number of medications and occasional notification issues, particularly when battery-saving modes are enabled on devices. In [15], the authors note that the system's dialogue recognition quality needs improvement to handle more complex user interactions. They also emphasize the limited features of the voice assistant, suggesting the need for advanced functionalities and better interaction capabilities. The authors in [18] point out challenges with technology familiarity among older adults, particularly those aged 75+ or from lower socioeconomic backgrounds, who may struggle to adopt digital tools. Privacy concerns regarding data usage, usability issues like voice recognition errors, and complex interfaces further hinder adoption. Additionally, the reliance on geriatrics experts rather than older adults limits the generalizability of the findings. They recommend addressing these issues with affordable, user-friendly designs, strong privacy measures, and studies involving diverse populations. The mobile application discussed in [19] also has limitations, including the repetitive nature of chatbot questions, which may lead to user boredom and reduced engagement over time. The system lacks gamification elements that could improve user motivation and interaction. While the chatbot effectively collects self-reported mental and physical states, the absence of sensor-based technologies restricts its ability to provide comprehensive health monitoring, such as detecting physical activities or environmental factors. Similarly, in [25], the proposed system is noted for high energy consumption due to continuous data collection and signal processing, which impacts device battery life. The system's reliance on MATLAB and a web-based platform limits its standalone functionality in real-world scenarios, while potential delays in signal transmission could affect its reliability during critical moments. The limitations identified in [26] include the challenge of sustaining long-term user motivation in gamified e-Health applications, as most tools demonstrate only short-

term engagement benefits. Additionally, the lack of user-centered design and limited involvement of healthcare professionals during development can result in applications that do not fully address the diverse needs of elderly users. The limitations in [27] include a lack of integration with robust behaviour change frameworks, over-reliance on extrinsic motivators, underutilization of intrinsic motivators, inconsistent terminology, and limited evaluation of long-term effectiveness. The [27] notes challenges such as balancing game complexity to avoid user frustration, ensuring long-term engagement, and addressing accessibility for elderly individuals unfamiliar with digital platforms. The long-term cognitive and emotional benefits of these apps remain uncertain, requiring further research.

2.6 Proposed Solutions

The proposed app addresses the limitations of existing systems by integrating AI-powered features, gamification, and robust emergency response tools to enhance the safety, cognitive engagement, and overall well-being of elderly users. One of the core functionalities is an AI chatbot designed to assist users with managing their medications and appointments. Using image recognition technology, the app ensures accurate recording of schedules and alerts caregivers to any updates or missed actions, ensuring seamless coordination between the elderly user and their support network. To promote cognitive health, the app offers brain-stimulating games such as puzzles, memory challenges, and matching tasks, all integrated with a ranking system to foster friendly competition and social interaction, keeping users mentally engaged. In the event of an emergency, the app uses sound detection to identify critical situations, such as shouting or falls. If the user does not respond to a prompt, the system automatically sends alerts, including calls and location details, to caregivers or emergency services. Additionally, the app features an offline emergency button, allowing users to access help even without an internet connection. To ensure accessibility, the app is designed with a simple interface, large buttons, and clear navigation, making it intuitive and user-friendly for elderly individuals, even those less familiar with technology. By combining these innovative features, the app provides a comprehensive, accessible, and reliable tool that enhances the independence, safety, and quality of life of elderly users while supporting caregivers in their roles.

CHAPTER 3

Project Scope and Objectives

3.1 Project Scope

This project aims to develop an **Elderly Companion Application** that leverages AI technologies and advanced features to enhance the safety, independence, and well-being of elderly users. The proposed system will integrate functionalities such as medication and appointment management, emergency detection, brain-stimulating games, and caregiver connectivity to create a comprehensive and user-friendly platform tailored for elderly individuals.

The application will feature an AI-powered chatbot to assist users in recording medication schedules and setting reminders for medical appointments. It will also include an image recognition feature that allows users to scan medicine labels and appointment notes, ensuring accurate data entry. Additionally, the chatbot will notify caregivers or relatives about any updates, such as missed medication or rescheduled appointments, and allow them to adjust if necessary.

To keep elderly users mentally engaged, the system will incorporate gamification through brain-stimulating games complete with a ranking system for friendly competition. For emergencies, the app will have a real-time detection feature capable of recognizing critical sounds like shouting and phone dropping. In such cases, the system will ask the user for confirmation and, if no response is received, automatically send emergency alerts, calls, and location details to caregivers or emergency services. A visible offline emergency button will also be included, allowing users to send an alert without requiring internet connectivity.

To address potential usability concerns, the system will be designed with a simple and intuitive interface, ensuring easy navigation for elderly users. Furthermore, the app

will prioritize energy efficiency and lightweight design to enable real-time responsiveness, minimizing waiting times.

The project will use advanced AI and machine learning techniques, such as Natural Language Processing (NLP) for chatbot interactions and sensor-based algorithms for emergency detection. The proposed Elderly Companion Application is expected to be completed by **October 2025**.

3.2 Project Objectives

The aim of the thesis is to develop a new mobile application that can help elderly users manage their daily health, safety, and cognitive engagement effectively. The project aims to achieve the following specific objectives:

1. **To develop an AI-powered chatbot** that assists elderly users in managing their medication schedules and appointment reminders.
2. **To enhance cognitive health** by incorporating gamified brain-stimulating activities with a ranking system to encourage engagement and social interaction.
3. **To promote caregiver connectivity** by sending notifications and updates regarding the user's medication schedule or emergency situations.
4. **To ensure user-friendly design and accessibility** by creating a simple interface with large buttons and intuitive navigation, tailored specifically for elderly users.

CHAPTER 4

Methods

The development of the Elderly Companion Application followed the **Rapid Application Development** (RAD) methodology, which emphasizes fast prototyping, iterative design, and incremental delivery. The processes were categorized into four distinct phases: Requirement Planning, Prototyping, Development and Integration, and Testing and Refinement.

4.2 Rapid Application Development method:

4.1.1 Requirement Planning:

This phase focused on identifying the key features of the application, such as medication and appointment management, emergency detection, and gamified brain activities. Tools and technologies like Android Studio, Firebase, and TensorFlow Lite were selected to meet the functional and technical requirements. The aim was to create a clear roadmap for the development process.

4.1.2 Prototyping:

In this phase, prototypes of core features were developed to test functionality and feasibility. Prototypes included an AI-powered chatbot for medication management, sensor-based emergency detection, and cognitive games for user engagement. These prototypes were internally tested and refined to ensure they met basic requirements and were ready for integration.

4.1.3 Development and Integration:

The refined prototypes were converted into fully functional modules and integrated into a single system. AI models for image recognition and emergency detection were trained, and Firebase was used for real-time data synchronization and storage. The user interface was designed to ensure accessibility and ease of use for elderly users.

4.1.4 Testing and Refinement:

The application underwent rigorous testing to ensure reliability and usability. Unit testing was performed for individual modules, and integration testing ensured all components worked seamlessly together. Feedback, if available, was incorporated to refine the application and address any issues.

4.2 System Design

4.2.1 Hardware

The hardware involved in this project includes a computer. The computer is used for processing and integrating AI technologies, such as the development of the chatbot, emergency detection system, and gamification features. It is also utilized for testing and refining the application during development.

Table 4.1 Specifications of laptop

Description	Specifications
Model	ROG Strix G713IH
Processor	AMD Ryzen 7 4800H with Radeon Graphics
Operating System	Windows 11 Home Single Language
Graphic	NVIDIA GeForce GTX 1650
Memory	16GB DDR4 RAM
Storage	512GB SATA HDD

4.2.2 Software

The software for this project will be developed using **Java**, a robust and widely used language for Android application development. The **Android Studio IDE** will be utilized to provide tools for coding, debugging, and testing the application. Java is chosen for its compatibility with a variety of Android devices, strong community support, and object-oriented features, which make it ideal for developing scalable and reliable mobile applications. For backend services, **Firebase** will be used for real-time data synchronization, user authentication, and secure data storage. **Firestore**, Firebase's NoSQL cloud database, will store critical data such as medication schedules, emergency alerts, and cognitive game progress. This ensures real-time

updates and seamless cross-session data synchronization. Firebase's adherence to GDPR and HIPAA standards ensures secure handling of sensitive user data. The application's AI-powered chatbot will be implemented using **Natural Language Processing (NLP)** frameworks compatible with Java, enabling it to manage medication schedules, answer user queries, and provide reminders. For image recognition, **TensorFlow Lite** will be integrated to process medication labels and appointment notes directly on the device, even offline. Emergency detection algorithms will utilize device sensors (microphone and accelerometer) to identify critical events like falls, shouting, or phone dropping in real-time. Cognitive engagement features, including brain-stimulating games, will be built as interactive Java-based activities. Firebase will store game progress and rankings, ensuring consistency and accessibility across user sessions. By combining Java's strong object-oriented capabilities with Firebase's backend and TensorFlow's AI features, the Elderly Companion Application will deliver a high-performing, accessible, and secure solution for elderly care.

4.3 System Workflow

4.3.1 Block Diagram of the Elderly Care Application

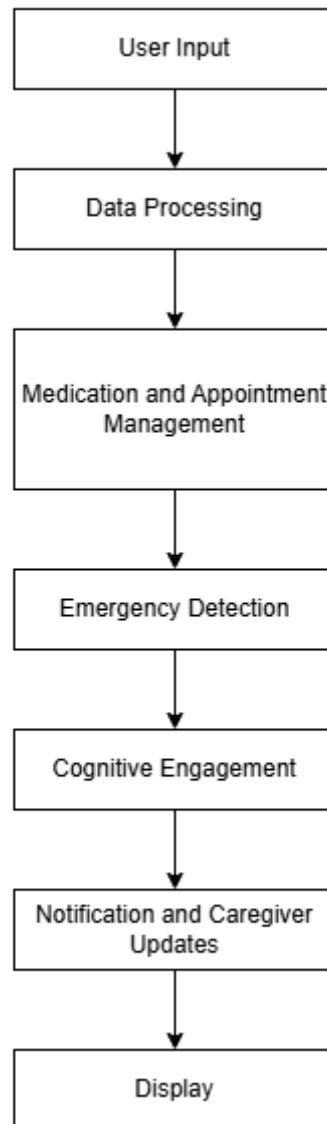


Figure 3.1 Block Diagram of the Elderly Care Application

The block diagram outlines the step-by-step workflow of the Elderly Companion Application, from the initial user input to the final output displayed on the application interface.

User Input:

The system begins with user interactions through the mobile application. Users can scan medication labels to record information, set reminders for medical appointments, or interact with the AI-powered chatbot for assistance. Additionally, emergency detection sensors continuously monitor for critical events such as falls, shouting, or the dropping of a phone. The accuracy of these inputs plays a significant role in the efficiency of the application.

Data Processing:

Once the user inputs are collected, the system processes the data using advanced technologies. For image recognition tasks, such as analysing scanned medication labels, the system leverages TensorFlow Lite models to identify and extract relevant details. Similarly, the AI chatbot processes natural language inputs using Natural Language Processing (NLP) frameworks to respond accurately and assist the user. The system also processes sensor data to analyse and detect anomalies related to emergencies.

Medication and Appointment Management:

The application handles all user data related to medications and appointments. It stores medication names, dosages, and timing schedules while managing reminders for upcoming appointments. Real-time synchronization is achieved using Firebase, ensuring that all data is securely stored and accessible across sessions. Users can modify their schedules as needed, and the system automatically updates and notifies caregivers or relatives if there are any changes.

Emergency Detection:

One of the core features of the application is its emergency detection system. Using sensors such as the device's microphone and accelerometer, the system monitors for unusual events like shouting, a fall, or a dropped phone. If a critical event is detected,

the system prompts the user to confirm their safety. In the absence of a response, it automatically sends emergency alerts, including the user's location, to caregivers or emergency services.

Cognitive Engagement:

To support cognitive health, the application includes interactive brain-stimulating games. These games, such as puzzles and matching activities, are designed to keep the user mentally active and engaged. The games are tailored to the needs of elderly users, offering varying levels of difficulty to cater to different cognitive abilities. Progress and scores are stored in Firebase, allowing users to track their achievements over time.

Notification and Caregiver Updates:

Notifications play a critical role in the application by keeping both the user and their caregivers informed. The system sends reminders for medications, alerts for missed doses, and updates in case of emergencies. Caregivers receive real-time notifications through Firebase Cloud Messaging (FCM), enabling them to monitor the user's health and safety remotely.

Display:

Finally, the system presents all outputs, such as medication schedules, emergency alerts, and game progress, on the application interface. The interface is designed with simplicity and accessibility in mind, featuring large buttons, clear navigation, and visually appealing layouts. All data is updated in real-time, ensuring that users and caregivers receive the most accurate and current information.

REFERENCES

Note: For the latest version of the Guide & Examples, please refer to UTAR library IEEE Referencing Style:

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- [1] World Health Organization (WHO). “Ageing and Health.” WHO. <https://www.who.int/news-room/fact-sheets/detail/ageing-and-health> (retrieved November 18, 2024).
- [2] Md. Isa, F. Noor, S. Wei, G. Wei, S.D. Syed Hussain, H. Mohamad Ibrahim, et al., "Exploring the facet of elderly care centre in multiethnic Malaysia", *PSU Research Review*, vol. 6, no. 1, pp. 17-38, 2020.
- [3] H. Arimo, H. Ito, T. Suzuki, A. Araki, T. Hosoi and M. Sawabe, "Reviewing the definition of “elderly”, *Geriatrics and Gerontology International*, vol. 6, no. 3, pp. 149-158, 2006.
- [4] F. Margret Sharmila, P. Tharun Kumar, G. Darvin Arockia Dass and S. Subramanian, "An Online Recruitment of Clinicians and Appointment of Patients using Mobile Application," *2021 Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV)*, Tirunelveli, India, 2021, pp. 1383-1385, doi: 10.1109/ICICV50876.2021.9388565.
- [5] J. Rodriguez. “Safety Issues for the Elderly Living Alone.” Griswold Home Care. <https://www.griswoldcare.com/blog/safety-issues-for-the-elderly-living-alone/> (retrieved November 18, 2024).
- [6] J. Doe. “5 Health Apps for Seniors.” Medical Alert Buyers Guide. <https://www.medicalalertbuyersguide.org/articles/5-health-apps-for-seniors/> (retrieved November 18, 2024).

REFERENCES

- [7] X. Zhao, L. Wang, C. Ge, X. Zhen, Z. Chen, J. Wang, et al., "Smartphone application training program improves smartphone usage competency and quality of life among the elderly in an elder university in China: A randomized controlled trial", *International Journal of Medical Informatics*, vol. 133, 2020.
- [8] AgeWiser, "Features," *AgeWiser*, 2024. [Online]. Available: <https://agewiser.ai/features/>. (retrieved November 22, 2024).
- [9] Medisafe, "Features," *Medisafe App*. [Online]. Available: <https://www.medisafe.com/>. (retrieved November 22, 2024).
- [10] CheckCharm, "Medisafe Medication Management App: In-Depth Review," *CheckCharm*. [Online]. Available: <https://www.checkcharm.com/reviews/medisafe-medication-management-app-in-depth-review>. (retrieved November 22, 2024).
- [11] MyTherapy, "Features and Benefits," *MyTherapy App*. [Online]. Available: <https://www.mytherapyapp.com/>. (retrieved November 22, 2024).
- [12] SteadyHealth, "MyTherapy Medication and Pill Reminder App: A Simple Therapy Reminder and Health Tracker," *SteadyHealth*. [Online]. Available: <https://www.steadyhealth.com/review/mytherapy>. (retrieved November 22, 2024).
- [13] Tech-Wonders, "MyTherapy Medication Reminder App Review," *Tech-Wonders*. [Online]. Available: <https://www.tech-wonders.com/2017/10/mytherapy-medication-reminder-app-review.html>. (retrieved November 22, 2024).

REFERENCES

- [14] S. Pakhmode, V. Poojary, P. Bhore, K. Thakur, and V. Dethé, "NLP based AI Voice Assistant," *INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT*, vol. 07, no. 03, Mar. 2023, doi: 10.55041/ijssrem18521.
- [15] S. Subhash, P. N. Srivatsa, S. Siddesh, A. Ullas, and B. Santhosh, "Artificial Intelligence-based Voice Assistant," *2020 Fourth World Conference on Smart Trends in Systems, Security and Sustainability (WorldS4)*, 2020, pp. 593-596, doi: 10.1109/WorldS450073.2020.9210344.
- [16] B. Setiaji and F. W. Wibowo, "Chatbot Using a Knowledge in Database: Human-to-Machine Conversation Modeling," *2016 7th International Conference on Intelligent Systems, Modelling and Simulation (ISMS)*, 2016, pp. 72–77. doi: 10.1109/ISMS.2016.53
- [17] S. Kumari, Z. Naikwadi, A. Akole, and P. Darshankar, "Enhancing College Chat Bot Assistant with the Help of Richer Human Computer Interaction and Speech Recognition," *Proceedings of the International Conference on Electronics and Sustainable Communication Systems (ICESC 2020)*, 2020, pp. 427–430. doi: 10.1109/ICESC.2020.9182777
- [18] M. Gudala, M. E. T. Ross, S. Mogalla, M. Lyons, P. Ramaswamy, and K. Roberts, "Benefits of, Barriers to, and Needs for an Artificial Intelligence–Powered Medication Information Voice Chatbot for Older Adults," *JMIR Aging*, vol. 5, no. 2, p. e32169, 2022. doi: 10.2196/32169

REFERENCES

- [19] H. Miura, M. Akiyama, and Y. Okazaki, "Assisting Personalized Healthcare of Elderly People: Developing a Rule-Based Virtual Caregiver System Using Mobile Chatbot," *Sensors*, vol. 22, no. 10, p. 3829, 2022. doi: 10.3390/s22103829
- [20] J. O. V. Paiva, R. M. C. Andrade, P. A. M. de Oliveira, P. Duarte, I. S. Santos, A. L. de P. Evangelista, R. L. Theophilo, L. O. M. de Andrade, and I. C. de H. C. Barreto, "Mobile applications for elderly healthcare: A systematic mapping," *PLoS ONE*, vol. 15, no. 7, p. e0236091, July 2020. doi: 10.1371/journal.pone.0236091
- [21] Z. Dahari, C. C. Jun, P. J. Ze, N. N. M. Zakir, M. N. F. M. Fauzi, M. H. S. M. Azam, and N. Hamiri, "Development of Smart Elderly Care Mobile Application for Health Management System," *2022 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET)*, 2022, pp. 1–6. doi: 10.1109/IICAIET55139.2022.9936853
- [22] World Health Organization, "Falls," *WHO*, 26 April 2021. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/falls>. (retrieved November 22, 2024).
- [23] T. Stampfler, M. Elgendi, R. R. Fletcher, and C. Menon, "Fall detection using accelerometer-based smartphones: Where do we go from here?" *Frontiers in Public Health*, vol. 10, 2022, Art. no. 996021, doi: 10.3389/fpubh.2022.996021
- [24] World Health Organization, *Global Report on Falls Prevention in Older Age*, Geneva, Switzerland: World Health Organization, 2007. [Online]. Available: <https://www.who.int/publications/i/item/9789241563536>. (retrieved November 22, 2024).

REFERENCES

- [25] A. Benba, M. Akki, and S. Sandabad, "The application of machine learning on the sensors of smartphones to detect falls in real-time," *IAPGOS*, vol. 2, pp. 50–55, 2023, doi: 10.35784/iapgos.3459
- [26] S. Sardi, A. Idri, J. Fernández-Alemán, and A. Toval, "A systematic review of gamification in e-Health," *Journal of Biomedical Informatics*, vol. 71, pp. 31–48, 2017. [Online]. Available: <https://doi.org/10.1016/j.jbi.2017.05.011>. (retrieved December 4, 2024).
- [27] S. Sardi, A. Idri, and J. L. Fernández-Alemán, "A systematic review of gamification in e-Health," *Journal of Biomedical Informatics*, vol. 71, pp. 31–48, 2017. [Online]. Available: <https://doi.org/10.1016/j.jbi.2017.05.011>. (retrieved December 4, 2024).
- [28] J. Siricharoen, "Creative Brain Training Games for Elderly," *CASA 2023 International Conference Proceedings*, 2023, pp. 105-112. [Online]. Available: <https://doi.org/10.1109/CASA2023.105>. (retrieved December 4, 2024).